Topic：Understanding Linear Regression: Impact of Experience on Salary with Performance Evaluation and Critical Analysis

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Course Title: Machine Learning and Neutral Networking

Introduction:

Machine Learning is a field of computer science that enables systems to learn patterns from data and make predictions. One of the most fundamental and widely used algorithms in supervised learning is Linear Regression. This tutorial focuses on analyzing the relationship between years of experience and salary using Linear Regression. In addition to explaining the basic concept, this tutorial evaluates model performance using appropriate metrics and discusses its assumptions, limitations, and real-world applications. The aim is to provide a clear and practical understanding so that beginners can apply this technique effectively in real-world scenarios.

Objective:

The objectives of this tutorial are:

- To understand the relationship between experience and salary
- To demonstrate how Linear Regression works mathematically and practically
- To evaluate model performance using MSE and R^2 score
- To critically analyze strengths and limitations of the model

Dataset Description:

This tutorial uses a dataset containing two variables:

Variable	Description	Type
Years Experience	Independent variable	Numeric
Salary	Dependent variable	Numeric

Why this dataset?

This dataset is simple and ideal for beginners. It clearly shows a linear relationship, making it perfect for demonstrating Linear Regression concepts. The dataset is **synthetic but realistic,** commonly used in ML tutorials for educational purposes.

Theory of Linear Regression:

Linear Regression models the relationship between variables using:

$$y = mx + b$$

Where:

y = predicted salary

x = years of experience

m = slope (rate of increase)

b = intercept

How it works?

The model finds the best-fit line by minimizing the error between predicted and actual values using **Least Squares Method.** The Least Squares Method minimizes the sum of squared residuals (the difference between actual and predicted values). This ensures that the regression line is the best possible fit for the given data.

Mathematically, the objective is to minimize:

$\sum(y_i - \hat{y}_i)^2$

This optimization makes Linear Regression highly efficient for prediction tasks involving continuous variables.

Assumptions of Linear Regression:

For accurate results, Linear Regression assumes:

Linearity: Relationship must be linear

Independence: Observations are independent

Homoscedasticity: Constant variance of errors

No multicollinearity: Inputs not highly correlated

Normal distribution of errors

Note: If these assumptions are violated, model performance decreases.

Overfitting and Underfitting:

Overfitting occurs when a model learns not only the underlying pattern but also the noise in the data, leading to poor performance on unseen data. Underfitting occurs when the model is too simple to capture the underlying trend.

In this tutorial, Linear Regression avoids overfitting due to its simplicity. However, it may underfit if the real-world relationship is non-linear.

Methodology:

Steps followed:

1. Dataset creation
2. Train-test split for model validation
3. Model training using Linear Regression
4. Prediction
5. Model evaluation (MSE & R^2 score)
6. Data visualization

The model is trained using training data and evaluated on unseen test data to ensure a fair and reliable assessment of performance.

Train-Test Split:

To ensure that the model generalizes well to unseen data, the dataset is divided into training and testing sets. The model is trained on the training data and evaluated on the testing data to avoid overfitting. The dataset is divided into training and testing sets using an 80-20 ratio.

Implementation (Python Code):

```
# Import Libraries

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

from sklearn.model_selection import train_test_split

# Create & Load Dataset

data = {

    "YearsExperience": [1.1,1.5,2.0,2.2,3.0,3.2,4.0,4.5,5.0,5.5,6.0,6.8,7.0,8.0,9.0,10.0],

    "Salary":
[39343,46205,43525,39891,56642,60150,55794,61111,66029,67938,81363,91738,93940,101302,105582,109431]
```

```
}
df = pd.DataFrame(data)
df.head()
# Features
X = df[['YearsExperience']]
y = df['Salary']
# Train- Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Model Training
model = LinearRegression()
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Evaluation
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("MSE:", mse)
print("R² Score:", r2)
# Data Visualization & Graphs
# 1. Linear Regression (Test Data)
plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred)
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
```

```
plt.title("Linear Regression (Test Data)")
plt.show()
# 2. Residual Plot
residuals = y_test - y_pred
plt.scatter(y_pred, residuals)
plt.axhline(y=0)
plt.xlabel("Predicted Values")
plt.ylabel("Residuals")
plt.title("Residual Plot")
```

The dataset was split into training and testing sets (80% training, 20% testing) to evaluate model performance on unseen data. This helps ensure that the model generalizes well and avoids overfitting. The dataset is split into training and testing sets using an 80-20 ratio. The model is trained on training data and predictions are made on test data.

Data Visualization:

1. **Scatter Plot for training and testing data + regression Line (Experience vs Salary):**

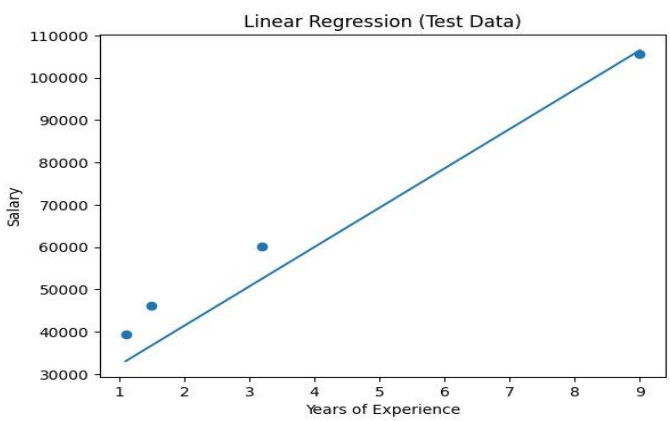

Figure 1: Linear relationship between experience and salary with regression line.

Observations:

- The plot shows a **linear regression** line modeling the relationship between **Years of Experience** (X-axis) and **Salary** (Y-axis).
- The regression line indicates a **positive linear correlation:** salary increases as experience increases.
- Points are close to the line, suggesting the linear model is a good fit for the data.

2. Residual Plot

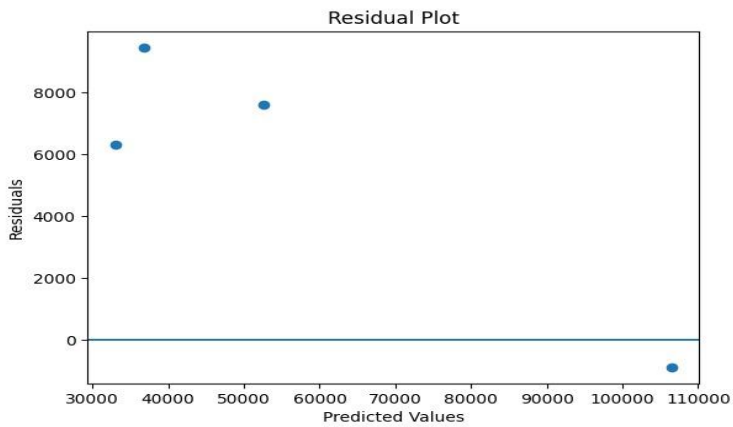

Figure 2: Relationship between predicted values and

Residuals with regression line.

Observations:

- Residuals represent the **difference** between actual salary values and the model's predicted values.
- Residuals are scattered randomly around the zero line, showing the model's predictions are unbiased and the errors are normally distributed.
- The absence of a pattern in residuals confirms the **linear regression** assumption is valid for this data.
- Residual plots help evaluate whether the assumptions of linear regression are satisfied.
- Random distribution of residuals around zero indicates a good model fit.

Results and Discussion:

R^2 Score ≈ High (~0.95+): Model explains most of the variance

MSE = Low: Predictions are close to actual values

Interpretation: This indicates that Linear Regression performs very well on this dataset.

The model achieves a high R^2 score and low mean squared error, indicating that it performs well on the given dataset. Since the evaluation is performed on unseen test data, the results are more reliable and better reflect real-world performance. However, due to the simplicity and limited size of the dataset, the model may not generalize perfectly to more complex real-world scenarios.

Critical Analysis:

Strengths:

- Simple and easy to implement
- Highly interpretable
- Works well with linear data

For improved performance, more features and advanced machine learning models can be explored in future work.

Limitations:

- Cannot handle non-linear relationships
- Sensitive to outliers
- Assumes perfect linearity

Additionally, Linear Regression assumes a strictly linear relationship, which is rarely the case in real-world scenarios. The model is also sensitive to outliers, which can significantly distort predictions.

Real-Life Applications:

- Salary prediction
- House price estimation
- Sales forecasting
- Business trend analysis

Personal Reflection:

While working on this tutorial, I learned that Linear Regression is a powerful yet simple tool for prediction tasks. However, I also realized that it is not suitable for complex datasets where relationships are non-linear. This helped me understand the importance of choosing the right model based on data. If I had more time, I would extend this project by applying Polynomial Regression and comparing its performance with Linear Regression to better understand non-linear patterns.

Ethical Considerations:

Machine learning models can introduce bias if the dataset is not fair. In salary prediction, bias may lead to discrimination. It is important to ensure fairness and transparency in model usage. In salary prediction, biased data (e.g., gender or regional bias) may lead to unfair outcomes. Therefore, fairness and transparency must be ensured when deploying machine learning models.

Accessibility:

This tutorial is designed to be accessible to a wide range of users. Simple and clear language is used so that beginners can easily understand the concepts. All graphs include proper labels and titles to improve readability. The visualizations are created using clear and distinguishable elements to ensure that they can be interpreted by users with different visual abilities. Additionally, the structure of the tutorial is organized with headings and sections, making it easy to navigate. The code is simple, well-structured, and can be executed step-by-step, allowing users with basic programming knowledge to follow along without difficulty.

Conclusion:

Linear Regression is a simple and effective technique for modeling the relationship between years of experience and salary. In this tutorial, the model was trained using training data and evaluated on unseen test data to ensure reliable performance. The results demonstrate that the model performs well when the assumptions of linear regression are satisfied. The visualizations confirm a strong positive linear relationship between experience and salary. Overall, Linear Regression is suitable for problems where relationships are approximately linear, while more advanced models may be required for complex datasets.

References:

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- Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning.

GitHub Repository:

https://github.com/nimrakhizar9822-droid/MLT-Assignment/blob/main/README.md